

Assignment No. : 02

Q.1) What is GitHub? What is version control in GitHub?

→ Github is a web-based platform used for version control and collective software development. It hosts Git repositories and provides tools for managing code, tracking changes, and working with teams.

Key features of GitHub:

- Git Repository Hosting: Stores code and tracks changes using Git.
- Collaborative: Allows multiple development to work on the same project.
- Pull Request: Used for reviewing and merging changes.
- Issue Tracking: Helps in tracking bugs and feature requests.

What is Version Control in GitHub?

Version control is a system that records changes to a file, or set of files over time so that you can recall specific versions later. Github uses Git, a distributed version control system, to manage and track code changes.

2) What is Angular?

→ Angular is a Typescript-based open source front-end framework developed by Google for building dynamic web application. It is mainly used to create Single Page Application where content update dynamically without reloading the page.

Key Features of Angular:

- Component-Based Architecture: Application are built using reusable components.
- Two way Data Binding: Automatically synchronizes data between the model and view.
- Built in HTTP Client: Allows communication with backend APIs.
- Routing Module: Enables navigation between pages.

3) What is Docker?

→ Docker is an open-source platform that allows developers to build, package, and distributed application inside containers. Containers are lightweight, standalone environments that include the application code along with its dependencies, ensuring it runs the same way in different environments.

Key Features of Docker:

- Portability: Works the same in development testing and production.
- Isolation: Each container runs independently with its own resources.
- Scalability: Easily scales application across multiple environments.
- Efficiency: Uses fewer resources compared to traditional virtual machines.

Q.4) How to push code on GitHub?

→ Steps to push code on GitHub:

1. Create a GitHub Repository:

- Go to GitHub, create a new Repository

2. Initialize Git in the Project Folder:
`git init.`

3. Connect to GitHub Repository:

`git remote add origin <repository-url>`

4. Stage & Commit changes:

`git add.`

`git commit -m "initial commit"`

5. Push code to GitHub:

`git push -u origin main`

Use master if your repo uses that instead of main

Now, code is successfully uploaded to GitHub.